Chapter 5

Jacobians:
Velocities and Static Forces

Overview

This chapter covers two methods for analyzing velocities and static forces:

- Velocity analysis of rigid bodies and velocity propagation
- Forward velocity problem and Jacobian
  - Joint directions and derivatives
  - Joint directions and cross products
  - Velocity transformation matrix and simpler Jacobians
  - Inverse velocity problem and singularities
  - Force transformation matrices and inverse static problem

5.1 Introduction

This chapter examines the notions of linear and angular velocities of rigid bodies.

The notion of differentiation of position vectors and rotation matrices will be used to define the velocity terms for general motion (i.e., a motion that includes both translation and rotation.

A systematic analysis of the velocities of the links of a manipulator will be conducted from the based o the end-effector (velocity propagation).

The study of both velocities and the static forces acting on a rigid body leads to a matrix entity called the **Jacobian**.

5.2 Notation – Differentiation of Position Vector

- The derivative of position vector $\boldsymbol{Q}$ with respect to frame $\{B\}$ is;

$$\frac{\mathrm{d}}{\mathrm{dt}}\,{}^{B}\boldsymbol{Q} = {}^{B}\boldsymbol{V}_{Q}$$

- The velocity vector can be expressed in terms of any other reference frame (such as frame $\{A\}$), using a leading superscript;

$${}^{A}({}^{B}\boldsymbol{V}_{Q}) = \frac{{}^{A}\mathrm{d}}{\mathrm{dt}}\,{}^{B}\boldsymbol{Q}$$

- Note that there is no need to express the outer leading superscript if the calculated velocity is written in terms of the frame of differentiation.

$${}^{B}({}^{B}\boldsymbol{V}_{Q}) = {}^{B}\boldsymbol{V}_{Q}$$

- Note that the outer leading superscript can be removed by explicitly including the rotation matrix that accomplishes the change in the reference frame.

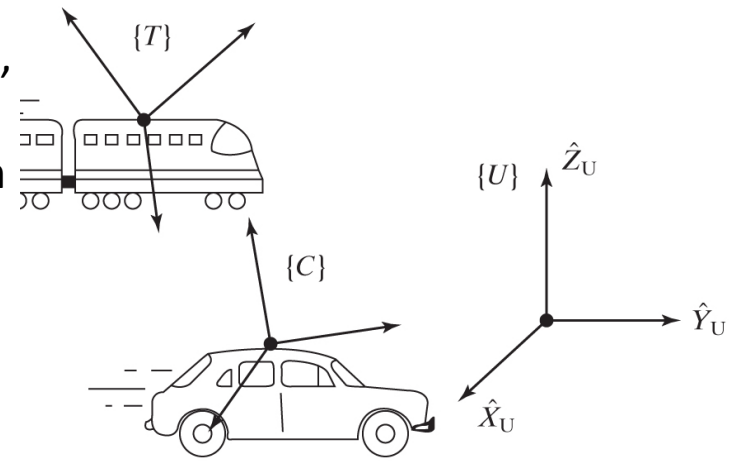
$${}^{A}({}^{B}\boldsymbol{V}_{Q}) = {}^{A}_{B}\boldsymbol{R}\,{}^{B}\boldsymbol{V}_{Q}$$

- The velocity of the origin of a frame (such as frame $\{C\}$) relative to universal frame $\{U\}$ can be expressed as:

$$\boldsymbol{v}_{C} = {}^{U}\boldsymbol{V}_{C_{ORG}}$$

5.2 Notation – Differentiation of Position Vector

<u>Example</u>: Figure below shows a fixed universe frame, $\{U\}$, a frame attached to a train traveling at 100 mph, $\{T\}$, and a frame attached to a car traveling at 30 mph, $\{C\}$. Both vehicles are leading in the $\hat{Y}$ direction of $\{U\}$. The rotation matrices, ${}^{U}_{T}\boldsymbol{R}$ and ${}^{U}_{C}\boldsymbol{R}$, are know and constant.



- Find $\frac{{}^{U}\mathrm{d}}{\mathrm{dt}}\,{}^{U}\boldsymbol{P}_{C_{ORG}}$

$$\frac{{}^{U}\mathrm{d}}{\mathrm{dt}}\,{}^{U}\boldsymbol{P}_{C_{ORG}} = {}^{U}\boldsymbol{V}_{C_{ORG}} = \boldsymbol{v}_C = 30\,\hat{Y}$$

- Find ${}^{C}({}^{U}\boldsymbol{V}_{T_{ORG}})$

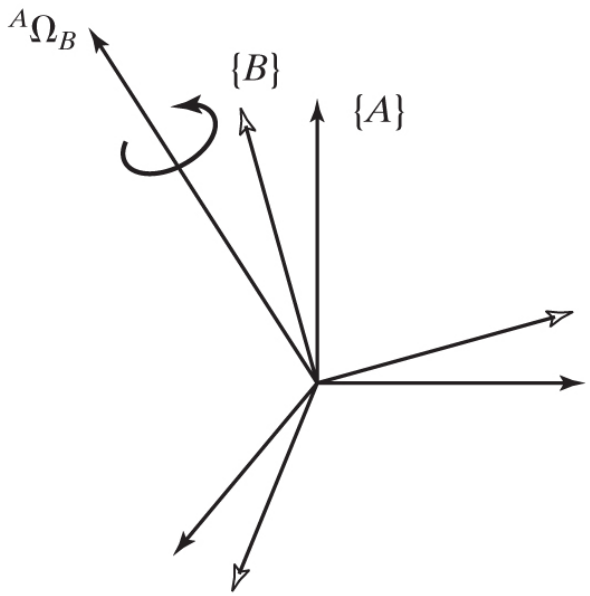
$${}^{C}\left({}^{U}\boldsymbol{V}_{T_{ORG}}\right) = {}^{C}\boldsymbol{v}_T = {}^{C}_{U}\boldsymbol{R}\boldsymbol{v}_T = {}^{C}_{U}\boldsymbol{R}\left(100\hat{Y}\right) = {}^{U}_{C}\boldsymbol{R}^{-1}\left(100\hat{Y}\right)$$

- Find ${}^{C}({}^{T}\boldsymbol{V}_{C_{ORG}})$

$${}^{C}\left({}^{T}\boldsymbol{V}_{C_{ORG}}\right) = {}^{C}_{T}\boldsymbol{R}\,{}^{T}\boldsymbol{V}_{C_{ORG}} = -{}^{U}_{C}\boldsymbol{R}^{-1}\,{}^{U}_{T}\boldsymbol{R}\left(70\hat{Y}\right)$$

5.2 Notation – The Angular Velocity Vector

- The angular velocity of frame $\{B\}$ relative to frame $\{A\}$ is the vector ${}^{A}\boldsymbol{\Omega}_{B}$

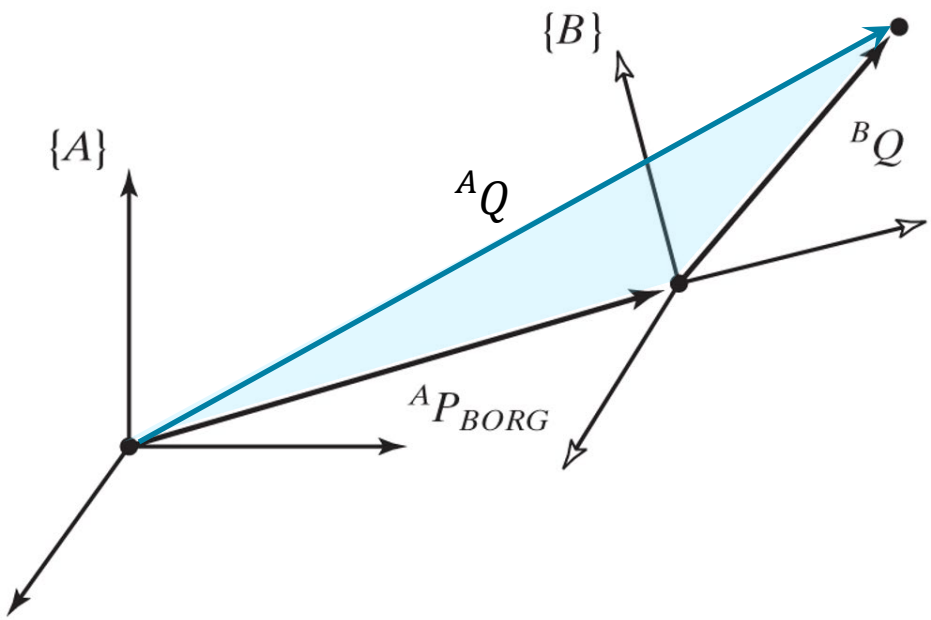


- For the angular velocity of frame $\{C\}$ relative to some understood reference $\{U\}$, a simplified notation is introduced as

$$\boldsymbol{\omega}_{C} = {}^{U}\boldsymbol{\Omega}_{C}$$

Linear Velocity

Assume the case of the general mapping of a position vector



The absolute velocity of point Q with respect to frame $\{A\}$ is found by differentiating the position vector with respect to time:

$$\frac{\mathrm{d}\left({}^{A}\boldsymbol{Q}\right)}{\mathrm{dt}} = \frac{\mathrm{d}\left({}^{A}\boldsymbol{P}_{B_{ORG}}\right)}{\mathrm{dt}} + \frac{\mathrm{d}\left({}^{A}_{B}\boldsymbol{R}\,{}^{B}\boldsymbol{Q}\right)}{\mathrm{dt}}$$

Linear Velocity – cont'd

If the orientation of frame $\{B\}$ relative to $\{A\}$ does not change, then then rotation matrix ${}^{A}_{B}\boldsymbol{R}$ remains constant. Hence,

$$\frac{\mathrm{d}\left({}^{A}\boldsymbol{Q}\right)}{\mathrm{dt}} = \frac{\mathrm{d}\left({}^{A}\boldsymbol{P}_{B_{ORG}}\right)}{\mathrm{dt}} + {}^{A}_{B}\boldsymbol{R}\,\frac{\mathrm{d}\left({}^{B}\boldsymbol{Q}\right)}{\mathrm{dt}}$$

$${}^{A}\boldsymbol{V}_{Q} = {}^{A}\boldsymbol{V}_{B_{ORG}} + {}^{A}_{B}\boldsymbol{R}\ {}^{B}\boldsymbol{V}_{Q}$$

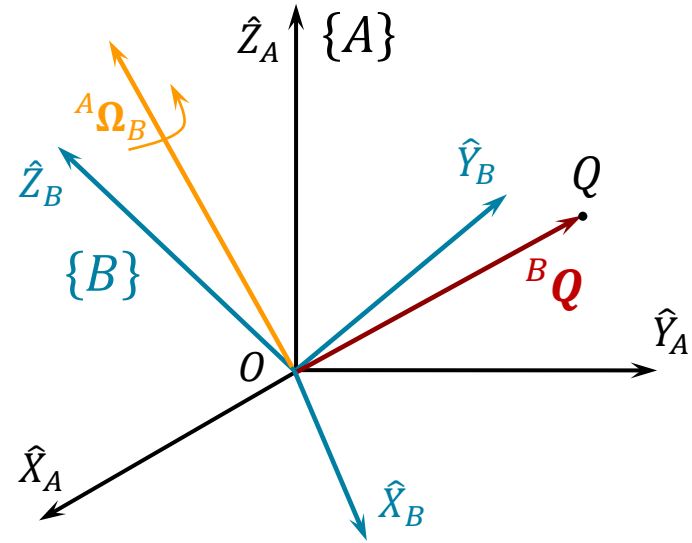
Absolute velocity of point Q relative to frame $\{A\}$

Absolute velocity of origin of $\{B\}$ relative to $\{A\}$

Velocity of Q with respect to $\{B\}$ expressed in $\{A\}$

Rotational Velocity

Assume the case of two reference frames that are coincident, and the moving reference frame rotates instantaneously about an axis of rotation with angular velocity of ${}^{A}\Omega_{B}$



$$ {}^{A}\boldsymbol{Q} = {}^{A}_{B}\boldsymbol{R}\ {}^{B}\boldsymbol{Q} $$

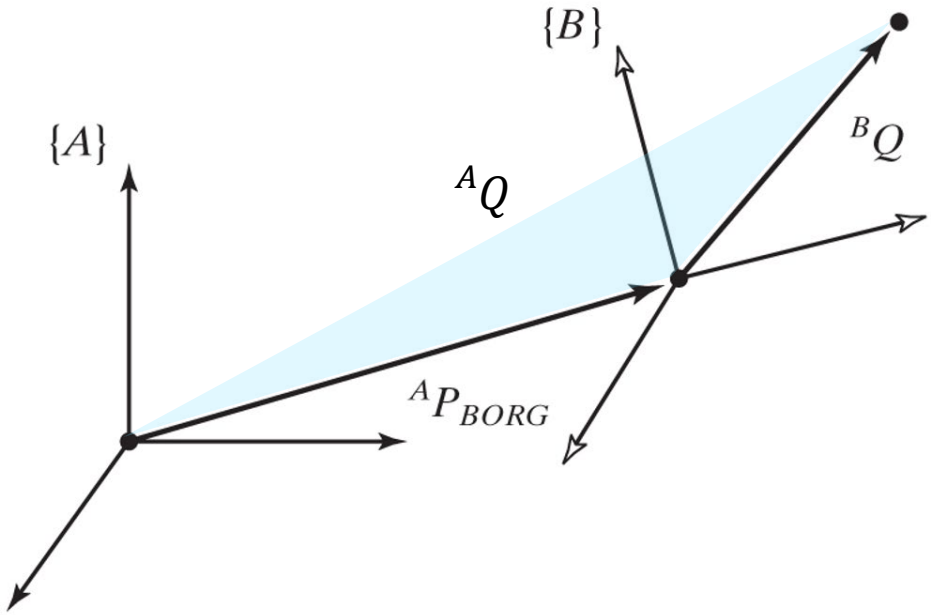
The velocity of point Q is given by

$$ \frac{\mathrm{d}({}^{A}\boldsymbol{Q})}{\mathrm{dt}} = {}^{A}_{B}\boldsymbol{R}\,\frac{\mathrm{d}({}^{B}\boldsymbol{Q})}{\mathrm{dt}} $$

$$ {}^{A}\boldsymbol{V}_{Q} = {}^{A}\boldsymbol{\Omega}_{B} \times {}^{A}_{B}\boldsymbol{R}\ {}^{B}\boldsymbol{Q} $$

General Velocity Representation

Assume the case of the general mapping of a position vector where the orientation between frames $\{A\}$ and $\{B\}$ changes over time:



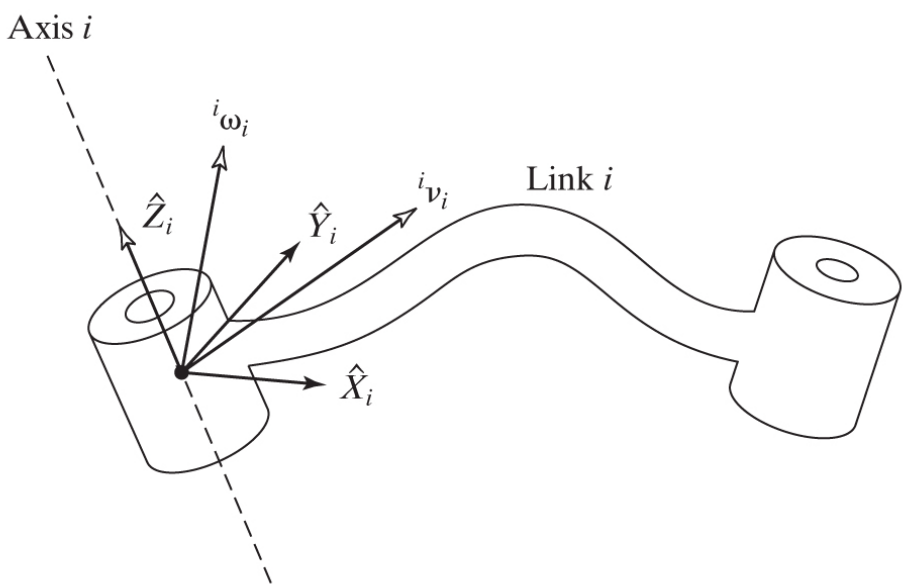
In the case of simultaneous linear and rotation velocity, the absolute velocity of Q with respect to frame $\{A\}$ is defined as

$$ {}^{A}\boldsymbol{V}_{Q} = {}^{A}\boldsymbol{V}_{B_{ORG}} + {}^{A}_{B}\boldsymbol{R}\ {}^{B}\boldsymbol{V}_{Q} + {}^{A}\boldsymbol{\Omega}_{B} \times {}^{A}_{B}\boldsymbol{R}\ {}^{B}\boldsymbol{Q} $$

5.5 Convention – Motion of the Links of a Robot

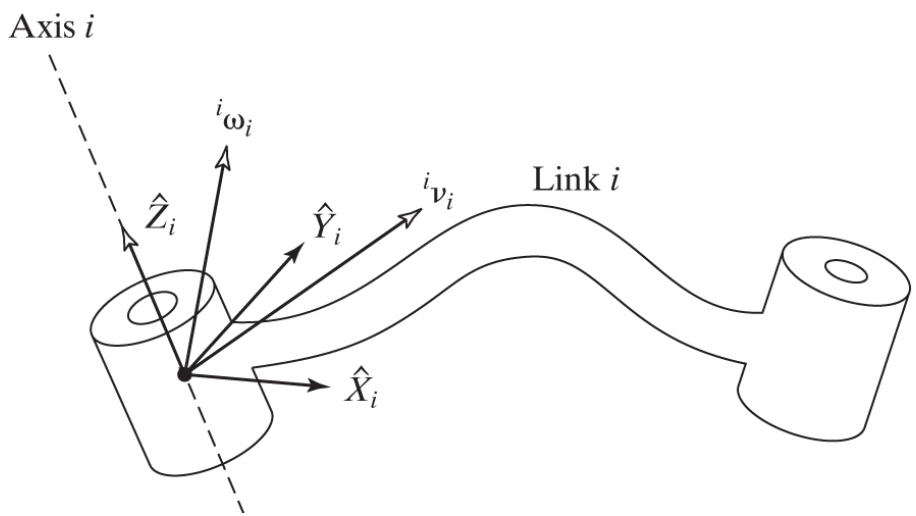
In considering the motion of robot links:

- Link frame $\{0\}$ is used as the reference frame
- v_i is the linear velocity of the origin of link frame $\{i\}$
- ω_i is the angular velocity of link frame $\{i\}$



5.6 Velocity Propagation from Link to Link

- Links undergo a change in position and orientation over time, i.e., linear and angular velocity.
- The velocity of each joint (joint rate) contributes to the linear and angular velocity of the links of the manipulator.
- The velocity of the links is a function of all the joint rates that precede the link. For example, the velocity of the end-effector is a function of all the joint rates. This characteristic is referred to as **velocity propagation**.
- Therefore, the linear and angular velocity of a link $i+1$ can be described as the velocity of link i plus the contribution of joint $i+1$, which precedes link $i+1$.

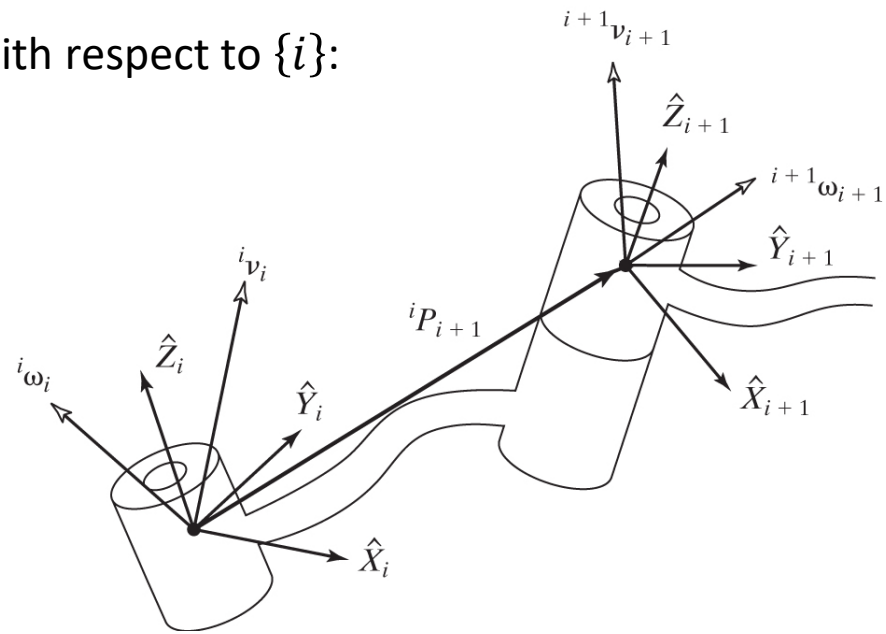


5.6 Velocity Propagation from Link to Link – Revolute Joints

Consider each link of the mechanism as a rigid body with linear and angular velocity vectors.

The **angular velocity** of link $i+1$ described with respect to $\{i\}$:

$$ {}^{i}\boldsymbol{\omega}_{i+1} = {}^{i}\boldsymbol{\omega}_{i} + {}^{i}_{i+1}\boldsymbol{R}\,\dot{\theta}_{i+1}\,{}^{i+1}\hat{Z}_{i+1} $$



where

$$ \dot{\theta}_{i+1}\,{}^{i+1}\hat{Z}_{i+1} = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_{i+1} \end{bmatrix} $$

Described with respect to $\{i+1\}$:

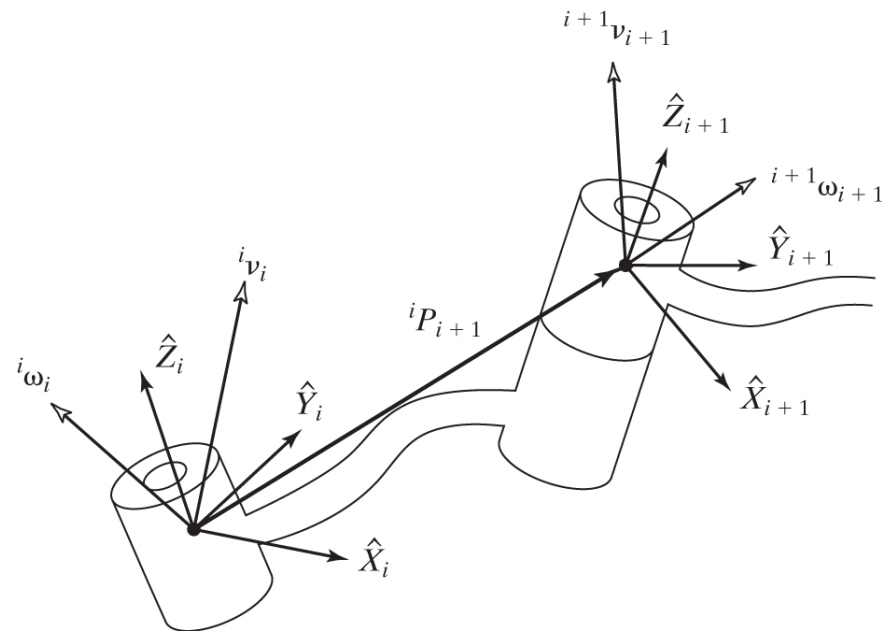
$$ ({}^{i+1}_{i}\boldsymbol{R})\ {}^{i}\boldsymbol{\omega}_{i+1} = ({}^{i+1}_{i}\boldsymbol{R})(\ {}^{i}\boldsymbol{\omega}_{i} + {}^{i}_{i+1}\boldsymbol{R}\,\dot{\theta}_{i+1}\,{}^{i+1}\hat{Z}_{i+1}) $$

$$ {}^{i+1}\boldsymbol{\omega}_{i+1} = {}^{i+1}_{i}\boldsymbol{R}\ \ {}^{i}\boldsymbol{\omega}_{i} + \dot{\theta}_{i+1}\,{}^{i+1}\hat{Z}_{i+1} $$

5.6 Velocity Propagation from Link to Link – Revolute Joints

The resulting **linear velocity** of frame $\{i+1\}$ is the same as that of the origin of frame $\{i\}$ plus the contribution made by the revolute joint:

$$ {}^{i}\boldsymbol{v}_{i+1} = {}^{i}\boldsymbol{v}_{i} + {}^{i}\boldsymbol{\omega}_{i} \times {}^{i}\boldsymbol{P}_{i+1} $$



Described with respect to $\{i+1\}$:

$$ {}^{i+1}_{i}\boldsymbol{R}\,({}^{i}\boldsymbol{v}_{i+1}) = {}^{i+1}_{i}\boldsymbol{R}\,({}^{i}\boldsymbol{v}_{i} + {}^{i}\boldsymbol{\omega}_{i} \times {}^{i}\boldsymbol{P}_{i+1}) $$

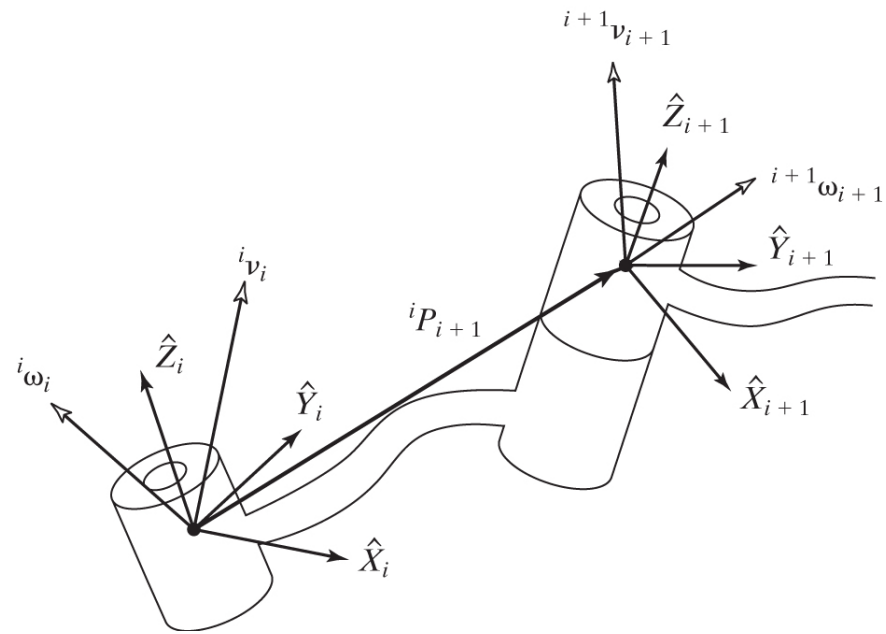
$$ {}^{i+1}\boldsymbol{v}_{i+1} = {}^{i+1}_{i}\boldsymbol{R}\,({}^{i}\boldsymbol{v}_{i} + {}^{i}\boldsymbol{\omega}_{i} \times {}^{i}\boldsymbol{P}_{i+1}) $$

5.6 Velocity Propagation from Link to Link – Prismatic Joints

Equivalent relationships can be achieved for prismatic joints:

The angular velocity of link $\{i+1\}$:

$$^{i+1}\boldsymbol{\omega}_{i+1} = {}^{i+1}_{i}\boldsymbol{R}\ \ {}^{i}\boldsymbol{\omega}_{i}$$



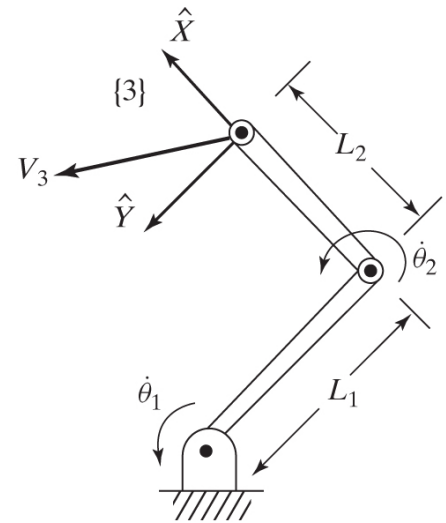
Note: there is no contribution of the prismatic joint to the angular velocity.

The linear velocity of the origin of $\{i+1\}$:

$$^{i+1}\boldsymbol{v}_{i+1} = {}^{i+1}_{i}\boldsymbol{R}\,(\,{}^{i}\boldsymbol{v}_{i} + {}^{i}\boldsymbol{\omega}_{i} \times {}^{i}\boldsymbol{P}_{i+1}) + \dot{d}_{i+1}\,{}^{i+1}\hat{Z}_{i+1}$$

5.6 Velocity Propagation from Link to Link

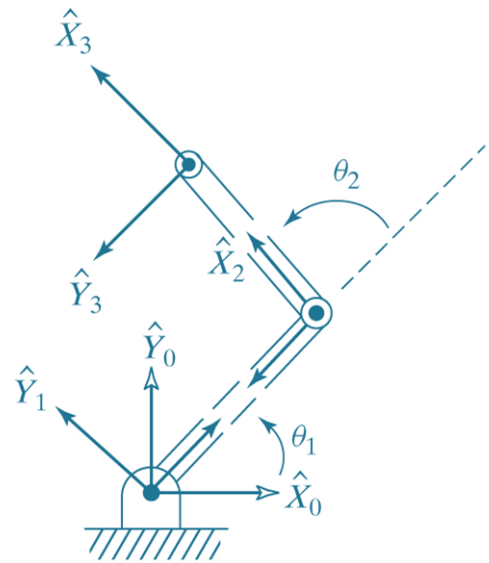
Example: A two-link manipulator with rotation joints is shown. Calculate the velocity of the tip of the arm as a function of joint rates. Give the answer in two form – in terms of frame {3}, and also in terms of frame {0}.



$$ {}^{0}_{1}\boldsymbol{T} = \begin{bmatrix} c1 & -s1 & 0 & 0 \\ s1 & c1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{1}_{2}\boldsymbol{T} = \begin{bmatrix} c2 & -s2 & 0 & L_1 \\ s2 & c2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{2}_{3}\boldsymbol{T} = \begin{bmatrix} 1 & 0 & 0 & L_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$



5.6 Velocity Propagation from Link to Link

Example: cont'd.

$$^{1}\boldsymbol{\omega}_1 = {}^{1}_{0}\boldsymbol{R}\ {}^{0}\boldsymbol{\omega}_0 + \dot{\theta}_1\ {}^{1}\hat{Z}_1 = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix}$$

$$^{1}\boldsymbol{v}_1 = {}^{1}_{0}\boldsymbol{R}\ (\ {}^{0}\boldsymbol{v}_0 + \ {}^{0}\boldsymbol{\omega}_0 \times \ {}^{0}\boldsymbol{P}_1) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$^{2}\boldsymbol{\omega}_2 = {}^{2}_{1}\boldsymbol{R}\ {}^{1}\boldsymbol{\omega}_1 + \dot{\theta}_2\ {}^{2}\hat{Z}_2 = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix}$$

$$^{2}\boldsymbol{v}_2 = {}^{2}_{1}\boldsymbol{R}\ (\ {}^{1}\boldsymbol{v}_1 + \ {}^{1}\boldsymbol{\omega}_1 \times \ {}^{1}\boldsymbol{P}_2) = \begin{bmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ L_1\dot{\theta}_1 \\ 0 \end{bmatrix} = \begin{bmatrix} L_1 s_2 \dot{\theta}_1 \\ L_1 c_2 \dot{\theta}_1 \\ 0 \end{bmatrix}$$

$$^{3}\boldsymbol{\omega}_3 = {}^{3}_{2}\boldsymbol{R}\ {}^{2}\boldsymbol{\omega}_2 + \dot{\theta}_3\ {}^{3}\hat{Z}_3 = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix}$$

$$^{3}\boldsymbol{v}_3 = {}^{3}_{2}\boldsymbol{R}\ (\ {}^{2}\boldsymbol{v}_2 + \ {}^{2}\boldsymbol{\omega}_2 \times \ {}^{2}\boldsymbol{P}_3) = \begin{bmatrix} L_1 s_2 \dot{\theta}_1 \\ L_1 c_2 \dot{\theta}_1 + L_2(\dot{\theta}_1 + \dot{\theta}_2) \\ 0 \end{bmatrix}$$

5.6 Velocity Propagation from Link to Link

Example: cont'd.

To find the velocities with respect to non-moving base frame, we rotate them with the rotation matrix, ${}^{0}_{3}\boldsymbol{R}$, which is:

$$ {}^{0}_{3}\boldsymbol{R} = {}^{0}_{1}\boldsymbol{R}\ {}^{1}_{2}\boldsymbol{R}\ {}^{2}_{3}\boldsymbol{R} = \begin{bmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} $$

The rotation yields:

$$ {}^{0}\boldsymbol{v}_3 = {}^{0}_{3}\boldsymbol{R}\ {}^{3}\boldsymbol{v}_3 = \begin{bmatrix} -L_1 s_1 \dot{\theta}_1 - L_2 s_{12}(\dot{\theta}_1 + \dot{\theta}_2) \\ L_1 c_1 \dot{\theta}_1 + L_2 c_{12}(\dot{\theta}_1 + \dot{\theta}_2) \\ 0 \end{bmatrix} $$

5.7 Jacobians & Forward Velocity Problem

The **Forward Velocity Problem** determines the velocity (linear an angular) of the end-effector ${}^{0}\boldsymbol{V}_{e.e.}$ as a function of the joint rates $\dot{\boldsymbol{q}}$.

A Jacobian is a mathematical concept that describes the variations of a given set of functions in a collective manner. It provides the linear mapping between time-varying variables.

The Jacobian allows the mapping between the angular/linear velocity of the end-effector as a function of the joint rates

Forward Velocity Problem

$$ {}^{0}\boldsymbol{V}_{e.e.} = {}^{0}\mathbf{J}_{e.e.}\dot{\boldsymbol{q}} \quad \text{where} \quad {}^{0}\boldsymbol{V}_{e.e.} = \begin{bmatrix} {}^{0}\boldsymbol{v}_{e.e.} \\ {}^{0}\boldsymbol{\omega}_{e.e.} \end{bmatrix} $$

Slide credit: Dr. F. Firmani

5.7 Jacobians & Forward Velocity Problem

In the previous example, if only the linear velocity is of interest, we can form the Jacobian matrix as:

$$
{}^{0}v_3 = \begin{bmatrix} -L_1 s_1 \dot{\theta}_1 - L_2 s_{12}(\dot{\theta}_1 + \dot{\theta}_2) \\ L_1 c_1 \dot{\theta}_1 + L_2 c_{12}(\dot{\theta}_1 + \dot{\theta}_2) \end{bmatrix} =_{2\times 1} \underbrace{\begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \end{bmatrix}}_{2\times 2\ {}^{0}J_3} \underbrace{\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}}_{2\times 1\ \dot{q}}
$$

$$
{}^{3}v_3 = \begin{bmatrix} L_2 s_2 \dot{\theta}_1 \\ L_1 c_2 \dot{\theta}_1 - L_2 c_{12}(\dot{\theta}_1 + \dot{\theta}_2) \end{bmatrix} =_{2\times 1} \underbrace{\begin{bmatrix} L_2 s_2 & 0 \\ L_1 c_2 + L_2 c_{12} & L_2 c_{12} \end{bmatrix}}_{2\times 2\ {}^{3}J_3} \underbrace{\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}}_{2\times 1\ \dot{q}}
$$

If we are interested in including angular velocity, the Jacobian matrix can be formed as:

$$
{}^{0}v_3 = \begin{bmatrix} -L_1 s_1 \dot{\theta}_1 - L_2 s_{12}(\dot{\theta}_1 + \dot{\theta}_2) \\ -L_1 c_1 \dot{\theta}_1 - L_2 c_{12}(\dot{\theta}_1 + \dot{\theta}_2) \\ \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \\ 1 & 1 \end{bmatrix}}_{3\times 2\ {}^{0}J_3} \underbrace{\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}}_{2\times 1\ \dot{q}}
$$

$$
{}^{3}v_3 = \begin{bmatrix} L_2 s_2 \dot{\theta}_1 \\ L_1 c_2 \dot{\theta}_1 - L_2 c_{12}(\dot{\theta}_1 + \dot{\theta}_2) \\ \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} L_2 s_2 & 0 \\ L_1 c_2 + L_2 c_{12} & L_2 c_{12} \\ 1 & 1 \end{bmatrix}}_{3\times 2\ {}^{3}J_3} \underbrace{\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}}_{2\times 1\ \dot{q}}
$$

5.7 Jacobians & Forward Velocity Problem

The Jacobian can be determined using two methods:

i. Joint Directions and Derivatives

ii. Joint Directions and Cross Products

5.7 Jacobians – Joint Directions and Derivatives

The **linear velocity** of the end-effector is determined by summing the partial derivatives of the position vector ${}^{0}\boldsymbol{P}_{ee}$ with respect to each joint variable times their joint rate,

$$ {}^{0}\boldsymbol{v}_{e.e.} = \sum_{i=1}^{n} \frac{\partial\, {}^{0}\boldsymbol{P}_{ee}}{\partial q_i} \dot{q}_i $$

The **angular velocity** of the end-effector is defined by the contribution of all the revolute joints,

$$ {}^{0}\boldsymbol{\omega}_{e.e.} = \sum_{i=1}^{n} \dot{\theta}_i \; {}^{0}\hat{Z}_i $$

5.7 Jacobians – Joint Directions and Derivatives

Thus, the Jacobian can be obtained as follows

$$
{}^{0}\mathbf{J}_{e.e.} = \begin{array}{c} \\ \\ \\ \end{array}
\begin{array}{c}
\begin{bmatrix} \dots & \dfrac{\partial\, {}^{0}\boldsymbol{P}_{ee}}{\partial \theta_i} & \dots & \dfrac{\partial\, {}^{0}\boldsymbol{P}_{ee}}{\partial d_j} & \dots \\ \dots & {}^{0}\hat{Z}_i & \dots & 0_{3\times 1} & \dots \end{bmatrix} \\
\underbrace{\qquad\qquad}_{\text{Revolute Joint } i} \quad \underbrace{\qquad\qquad}_{\text{Prismatic Joint } j}
\end{array}
\begin{array}{l} \\ \text{Linear Velocity} \\ \\ \text{Angular Velocity} \end{array}
$$

- The Jacobian is formed by sequentially adding a column that represents the corresponding joint (revolute or prismatic).
- The joint direction ${}^{0}\hat{Z}_i$ can be obtained by extracting the third column to the homogeneous transforms ${}^{0}_{i}\boldsymbol{T}$.
- The position vector ${}^{0}\boldsymbol{P}_{ee}$ is extracted from the last column of ${}^{0}_{ee}\boldsymbol{T}$.

5.7 Jacobians & Forward Velocity Problem

We can apply this method in the previous example:

$$ {}^{0}\mathbf{J}_3 = \begin{bmatrix} \dfrac{\partial\, {}^{0}\boldsymbol{P}_3}{\partial \theta_1} & \dfrac{\partial\, {}^{0}\boldsymbol{P}_3}{\partial \theta_2} \\ {}^{0}\hat{Z}_1 & {}^{0}\hat{Z}_2 \end{bmatrix} $$

We have

$$ {}^{0}_{1}\boldsymbol{T} = \begin{bmatrix} c1 & -s1 & 0 & 0 \\ s1 & c1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^{1}_{2}\boldsymbol{T} = \begin{bmatrix} c2 & -s2 & 0 & L_1 \\ s2 & c2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^{2}_{3}\boldsymbol{T} = \begin{bmatrix} 1 & 0 & 0 & L_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

We can calculate

$$ {}^{0}_{2}\boldsymbol{T} = \begin{bmatrix} c12 & -s12 & 0 & L_1 c1 \\ s12 & c12 & 0 & L_1 s1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^{0}_{3}\boldsymbol{T} = \begin{bmatrix} c12 & -s12 & 0 & L_1 c1 + L_2 c12 \\ s12 & c12 & 0 & L_1 s1 + L_2 s12 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

5.7 Jacobians & Forward Velocity Problem

$$ {}^{0}\boldsymbol{P}_{\mathbf{3}} = \begin{bmatrix} L_1 c1 + L_2 c12 \\ L_1 s1 + L_2 s12 \\ 0 \end{bmatrix} $$

$$ \frac{\partial\ {}^{0}\boldsymbol{P}_3}{\partial \theta_1} = \begin{bmatrix} -L_1 s1 - L_2 s12 \\ L_1 c1 + L_2 c12 \\ 0 \end{bmatrix} $$

$$ \frac{\partial\ {}^{0}\boldsymbol{P}_3}{\partial \theta_2} = \begin{bmatrix} -L_2 s12 \\ L_1 c12 \\ 0 \end{bmatrix} $$

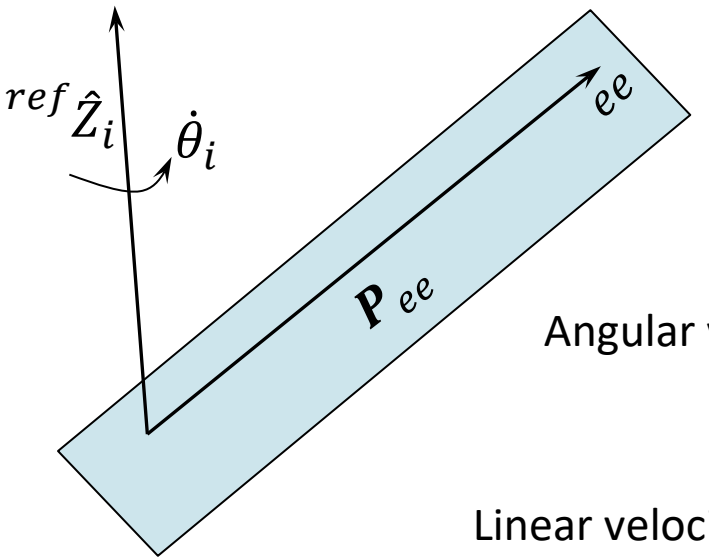
and

$$ {}^{0}\hat{Z}_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \qquad {}^{0}\hat{Z}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} $$

$$ {}^{0}\mathbf{J}_3 = \begin{bmatrix} \frac{\partial\ {}^{0}\boldsymbol{P}_3}{\partial \theta_1} & \frac{\partial\ {}^{0}\boldsymbol{P}_3}{\partial \theta_2} \\ {}^{0}\hat{Z}_1 & {}^{0}\hat{Z}_{\mathbf{2}} \end{bmatrix} = \begin{bmatrix} -L_1 s1 - L_2 s12 & -L_2 s12 \\ L_1 c1 + L_2 c12 & L_1 c12 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} $$

5.7 Jacobians – Joint Directions and Cross-products

Revolute Joints consider a rigid body rotating about an axis $\hat{Z}_i$ with an angular velocity $\dot{\theta}_i$. If the end-effector (ee) is a point on this rigid body, the contribution of each revolute joint to the velocity of ee can be determined as follows:



Angular velocity: $^{ref}\boldsymbol{\omega}_i = \dot{\theta}_i\, {}^{ref}\hat{Z}_i$

Linear velocity: $^{ref}\boldsymbol{v}_i = \dot{\theta}_i \left({}^{ref}\hat{Z}_i \times \boldsymbol{P}_{ee} \right)$

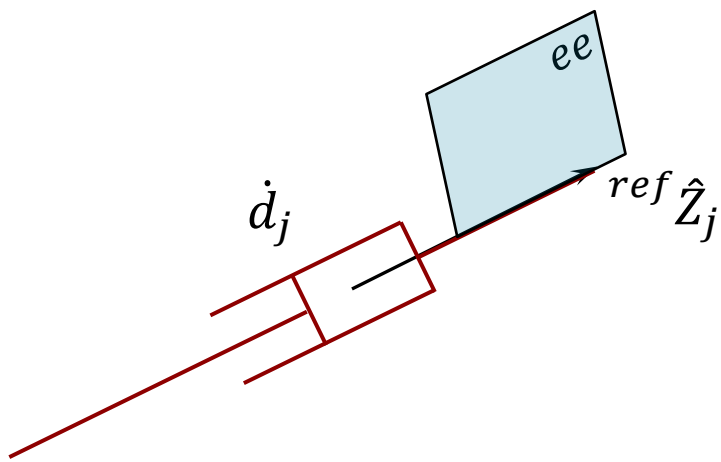
<u>*Note*</u>: Frame ref can be any reference frame, for example, $\{ref\} = \{0\}$.

5.7 Jacobians – Joint Directions and Cross-products

Prismatic Joints consider a rigid body attached to a prismatic joint sliding along axis $\hat{Z}_j$ with a velocity $\dot{d}_j$. While the contribution of this joint to the angular velocity of the rigid body is none, its contribution to the linear velocity of the rigid body is $\dot{d}_j$ along $\hat{Z}_j$:

Angular velocity: ${}^{ref}\boldsymbol{\omega}_j = 0_{3\times1}$

Linear velocity: ${}^{ref}\boldsymbol{v}_j = \dot{d}_j\,{}^{ref}\hat{Z}_j$



5.7 Jacobians – Joint Directions and Cross-products

Let $\{ref\} = \{0\}$, thus the Jacobian can be obtained as follows;

$$
{}^{0}\mathbf{J}_{e.e.} = \begin{bmatrix} \cdots & \underbrace{\begin{matrix} {}^{0}\hat{Z}_i \times {}^{0}\boldsymbol{P}_{i\to ee} \\ {}^{0}\hat{Z}_i \end{matrix}}_{\text{Revolute Joint } i} & \cdots & \underbrace{\begin{matrix} {}^{0}\hat{Z}_j \\ 0_{3\times 1} \end{matrix}}_{\text{Prismatic Joint } j} & \cdots \end{bmatrix} \begin{matrix} \text{Linear Velocity} \\ \text{Angular Velocity} \end{matrix}
$$

- The Jacobian is formed by sequentially adding a column that represents the corresponding joint (revolute or prismatic).
- The joint direction ${}^{0}\hat{Z}_i$ can be obtained from ${}^{0}_{i}\boldsymbol{T}$.
- The position vector ${}^{0}\boldsymbol{P}_{ee}$ can be found as

$$
{}^{0}\boldsymbol{P}_{i\to ee} = {}^{0}\boldsymbol{P}_{ee} - {}^{0}\boldsymbol{P}_{i} \quad \text{or} \quad {}^{0}\boldsymbol{P}_{i\to ee} = {}^{0}_{i}\boldsymbol{R}\ {}^{i}\boldsymbol{P}_{ee}
$$

5.7 Jacobians & Forward Velocity Problem

We can apply this method in the previous example:

$$ {}^{0}\mathbf{J}_3 = \begin{bmatrix} {}^{0}\hat{Z}_1 \times {}^{0}\boldsymbol{P}_{1\to 3} & {}^{0}\hat{Z}_2 \times {}^{0}\boldsymbol{P}_{2\to 3} \\ {}^{0}\hat{Z}_1 & {}^{0}\hat{Z}_2 \end{bmatrix} $$

From case 1

$$ {}^{0}_{1}\boldsymbol{T} = \begin{bmatrix} c1 & -s1 & 0 & 0 \\ s1 & c1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad {}^{0}_{2}\boldsymbol{T} = \begin{bmatrix} c12 & -s12 & 0 & L_1 c1 \\ s12 & c12 & 0 & L_1 s1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{0}_{3}\boldsymbol{T} = \begin{bmatrix} c12 & -s12 & 0 & L_1 c1 + L_2 c12 \\ s12 & c12 & 0 & L_1 s1 + L_2 s12 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Therefore

$$ {}^{0}\boldsymbol{P}_3 = \begin{bmatrix} L_1 c1 + L_2 c12 \\ L_1 s1 + L_2 s12 \\ 0 \end{bmatrix} \qquad {}^{0}\boldsymbol{P}_2 = \begin{bmatrix} L_1 c1 \\ L_1 s1 \\ 0 \end{bmatrix} \qquad {}^{0}\boldsymbol{P}_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} $$

5.7 Jacobians & Forward Velocity Problem

$$ {}^{0}\boldsymbol{P}_{1\to 3} = {}^{0}\boldsymbol{P}_{3} - {}^{0}\boldsymbol{P}_{1} = \begin{bmatrix} L_1 c1 + L_2 c12 \\ L_1 s1 + L_2 s12 \\ 0 \end{bmatrix} $$

$$ {}^{0}\boldsymbol{P}_{2\to 3} = {}^{0}\boldsymbol{P}_{3} - {}^{0}\boldsymbol{P}_{2} = \begin{bmatrix} L_1 c1 + L_2 c12 \\ L_1 s1 + L_2 s12 \\ 0 \end{bmatrix} - \begin{bmatrix} L_1 c1 \\ L_1 s1 \\ 0 \end{bmatrix} = \begin{bmatrix} L_2 c12 \\ L_2 s12 \\ 0 \end{bmatrix} $$

and

$$ {}^{0}\hat{Z}_1 \times {}^{0}\boldsymbol{P}_{1\to 3} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times \begin{bmatrix} L_1 c1 + L_2 c12 \\ L_1 s1 + L_2 s12 \\ 0 \end{bmatrix} = \begin{bmatrix} -L_1 s1 - L_2 s12 \\ L_1 c1 + L_2 c12 \\ 0 \end{bmatrix} $$

$$ {}^{0}\hat{Z}_2 \times {}^{0}\boldsymbol{P}_{2\to 3} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times \begin{bmatrix} L_2 c12 \\ L_2 s12 \\ 0 \end{bmatrix} = \begin{bmatrix} -L_2 s12 \\ L_2 c12 \\ 0 \end{bmatrix} $$

5.7 Jacobians & Forward Velocity Problem

$$
{}^{0}\mathbf{J}_3 = \begin{bmatrix} {}^{0}\hat{Z}_1 \times {}^{0}\boldsymbol{P}_{1\to 3} & {}^{0}\hat{Z}_2 \times {}^{0}\boldsymbol{P}_{2\to 3} \\ {}^{0}\hat{Z}_1 & {}^{0}\hat{Z}_2 \end{bmatrix} = \begin{bmatrix} -L_1 s1 - L_2 s12 & -L_2 s12 \\ L_1 c1 + L_2 c12 & L_1 c12 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}
$$

5.7 Jacobians – Remarks

Finding the Jacobian of a point at the tip of the end-effector can be tedious and difficult. Determining the singularities of this Jacobian involves computing the determinant of a 6 × 6 matrix!

As an alternative, since the end-effector link is a rigid body, one could determine the Jacobian with respect to the wrist, ${}^{0}\mathbf{J}_{w}$

$$ {}^{0}\boldsymbol{V}_{w} = {}^{0}\mathbf{J}_{w}\dot{\boldsymbol{q}} \qquad \text{where} \qquad {}^{0}\mathbf{J}_{w} = \begin{bmatrix} A_{3\times3} & 0_{3\times3} \\ B_{3\times3} & C_{3\times3} \end{bmatrix} $$

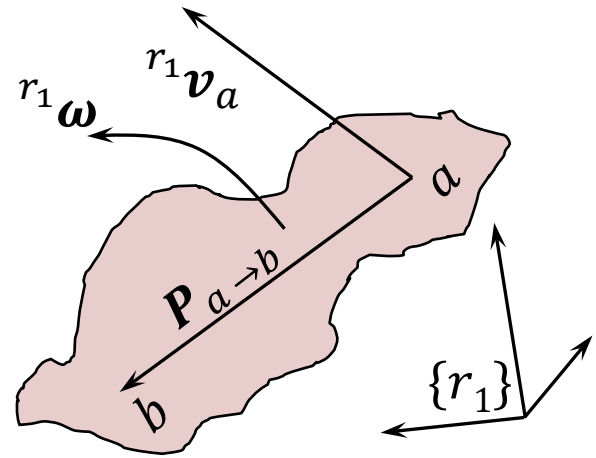
resulting in a much simpler determinant $\left| {}^{0}\mathbf{J}_{w} \right| = |A_{3\times3}||C_{3\times3}|$.

Once ${}^{0}\boldsymbol{V}_{w}$ is found, then a **velocity transformation** can be employed to find the velocity with respect to end-effector, ${}^{0}\boldsymbol{V}_{ee}$.

5.7 Jacobians – Remarks – Velocity Transformation $\boldsymbol{Tv}_{b\to a}$

Velocity Transformation $\boldsymbol{Tv}_{b\to a}$

Assume two points "a" and "b" on the same rigid body, let the velocity of point a be known with respect to a reference frame $\{r_1\}$. What is the velocity of point b with respect to frame $\{r_1\}$?



Known: $^{r_1}\boldsymbol{V}_a = \begin{bmatrix} ^{r_1}\boldsymbol{v}_a \\ ^{r_1}\boldsymbol{\omega} \end{bmatrix}$, find $^{r_1}\boldsymbol{V}_b = \begin{bmatrix} ^{r_1}\boldsymbol{v}_b \\ ^{r_1}\boldsymbol{\omega} \end{bmatrix}$

$$^{r_1}\boldsymbol{v}_b = {}^{r_1}\boldsymbol{v}_a + {}^{r_1}\boldsymbol{\omega} \times {}^{r_1}\boldsymbol{P}_{a\to b}$$

$$= {}^{r_1}\boldsymbol{v}_a + {}^{r_1}\boldsymbol{P}_{b\to a} \times {}^{r_1}\boldsymbol{\omega} = {}^{r_1}\boldsymbol{v}_a + {}^{r_1}\widetilde{\boldsymbol{P}}_{b\to a}\,{}^{r_1}\boldsymbol{\omega}$$

where the skew matrix $^{r_1}\widetilde{\boldsymbol{P}}_{b\to a}$ is

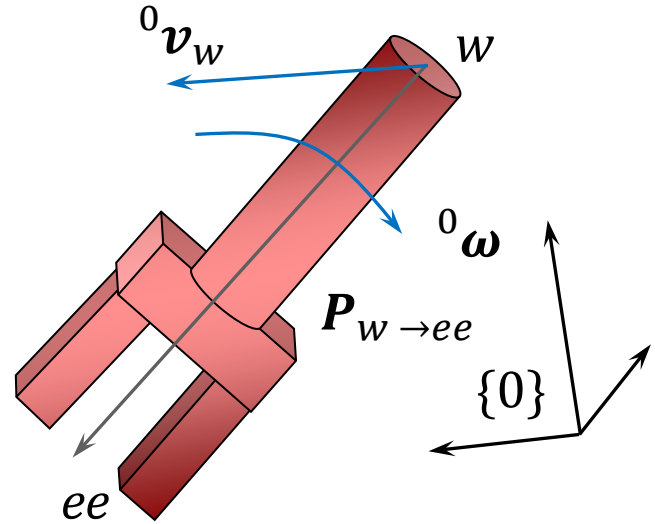
$$^{r_1}\widetilde{\mathbf{P}}_{b\to a} = \begin{bmatrix} 0 & -P_{b\to a_z} & P_{b\to a_y} \\ P_{b\to a_z} & 0 & -P_{b\to a_x} \\ -P_{b\to a_y} & P_{b\to a_x} & 0 \end{bmatrix}$$

$$\begin{bmatrix} ^{r_1}\boldsymbol{v}_b \\ ^{r_1}\boldsymbol{\omega} \end{bmatrix} = \underbrace{\begin{bmatrix} \boldsymbol{I}_{3\times3} & ^{r_1}\widetilde{\boldsymbol{P}}_{b\to a} \\ \mathbf{0}_{3\times3} & \boldsymbol{I}_{3\times3} \end{bmatrix}}_{\boldsymbol{Tv}_{b\to a}} \begin{bmatrix} ^{r_1}\boldsymbol{v}_a \\ ^{r_1}\boldsymbol{\omega} \end{bmatrix}$$

Velocity Transformation Matrix

5.7 Jacobians – Remarks – Velocity Transformation $\boldsymbol{Tv}_{ee\to w}$

Let ${}^{ref}\mathbf{J}_w$ be known (from slide 25) for a given set of joint rates $\dot{\boldsymbol{q}}$. The velocity of the wrist can be found. The general velocity transformation can be implemented to the end-effector to find ${}^{0}\boldsymbol{V}_{ee}$.



Known: ${}^{0}\boldsymbol{V}_w = \begin{bmatrix} {}^{0}\boldsymbol{v}_w \\ {}^{0}\boldsymbol{\omega} \end{bmatrix} = {}^{0}\mathbf{J}_w\dot{\boldsymbol{q}}$, find ${}^{0}\boldsymbol{V}_{ee} = \begin{bmatrix} {}^{0}\boldsymbol{v}_{ee} \\ {}^{0}\boldsymbol{\omega} \end{bmatrix}$

$$
\begin{aligned}
{}^{0}\boldsymbol{v}_{ee} &= {}^{0}\boldsymbol{v}_w + {}^{0}\boldsymbol{\omega} \times {}^{0}\boldsymbol{P}_{w\to ee} \\
&= {}^{0}\boldsymbol{v}_w + {}^{0}\boldsymbol{P}_{ee\to w} \times {}^{0}\boldsymbol{\omega} \\
&= {}^{0}\boldsymbol{v}_w + {}^{0}\widetilde{\boldsymbol{P}}_{ee\to w}\,{}^{0}\boldsymbol{\omega}
\end{aligned}
$$

where the skew matrix ${}^{0}\widetilde{\boldsymbol{P}}_{ee\to w}$ is: ${}^{0}\widetilde{\boldsymbol{P}}_{ee\to w} = \begin{bmatrix} 0 & -P_{ee\to w_z} & P_{ee\to w_y} \\ P_{ee\to w_z} & 0 & -P_{ee\to w_x} \\ -P_{ee\to w_y} & P_{ee\to w_x} & 0 \end{bmatrix}$

$$
\begin{bmatrix} {}^{0}\boldsymbol{v}_{ee} \\ {}^{0}\boldsymbol{\omega} \end{bmatrix} = \underbrace{\begin{bmatrix} \boldsymbol{I}_{3\times3} & {}^{0}\widetilde{\boldsymbol{P}}_{ee\to w} \\ \mathbf{0}_{3\times3} & \boldsymbol{I}_{3\times3} \end{bmatrix}}_{\boldsymbol{Tv}_{ee\to w}} \begin{bmatrix} {}^{0}\boldsymbol{v}_w \\ {}^{0}\boldsymbol{\omega} \end{bmatrix}
$$

Velocity Transformation Matrix

5.7 Jacobians – Remarks (${}^{0}\mathbf{J}_{w}$)

Joint Directions and Derivatives

$$
{}^{0}\mathbf{J}_{w} = \begin{bmatrix} \underbrace{\begin{matrix} \dfrac{\partial\, {}^{0}\boldsymbol{P}_{w}}{\partial \theta_i} \\ \\ {}^{0}\hat{Z}_i \end{matrix}}_{\text{Revolute Joint } i} & \cdots & \underbrace{\begin{matrix} \dfrac{\partial\, {}^{0}\boldsymbol{P}_{w}}{\partial d_j} \\ \\ 0_{3\times 1} \end{matrix}}_{\text{Prismatic Joint } j} & \cdots \end{bmatrix} \begin{matrix} \text{Linear Velocity} \\ \\ \text{Angular Velocity} \end{matrix}
$$

Joint Directions and Cross-Products

$$
{}^{0}\mathbf{J}_{w} = \begin{bmatrix} \cdots & \underbrace{\begin{matrix} {}^{0}\hat{Z}_i \times {}^{0}\boldsymbol{P}_{i\to w} \\ \\ {}^{0}\hat{Z}_i \end{matrix}}_{\text{Revolute Joint } i} & \cdots & \underbrace{\begin{matrix} {}^{0}\hat{Z}_j \\ \\ 0_{3\times 1} \end{matrix}}_{\text{Prismatic Joint } j} & \cdots \end{bmatrix} \begin{matrix} \text{Linear Velocity} \\ \\ \text{Angular Velocity} \end{matrix}
$$

Jacobian (${}^{0}\mathbf{J}_{w}$) Remarks

In order to evaluate the velocity of the end-effector based on a set of joint rates, two matrices were required: the Jacobian ${}^{0}\mathbf{J}_{w}$ and the velocity transformation matrix $\boldsymbol{Tv}_{ee\,\to w}$:

$$ {}^{0}\boldsymbol{V}_{ee} = \boldsymbol{Tv}_{ee\,\to w}\ {}^{0}\mathbf{J}_{w}\,\dot{\boldsymbol{q}} $$

The evaluation of ${}^{0}\mathbf{J}_{w}$ is much easier than finding ${}^{0}\mathbf{J}_{\mathrm{ee}}$. Also, for analyzing singularities, solving for the determinant is simpler;

$$ \left|\,{}^{0}\mathbf{J}_{w}\right| = |A_{3\times3}||C_{3\times3}| $$

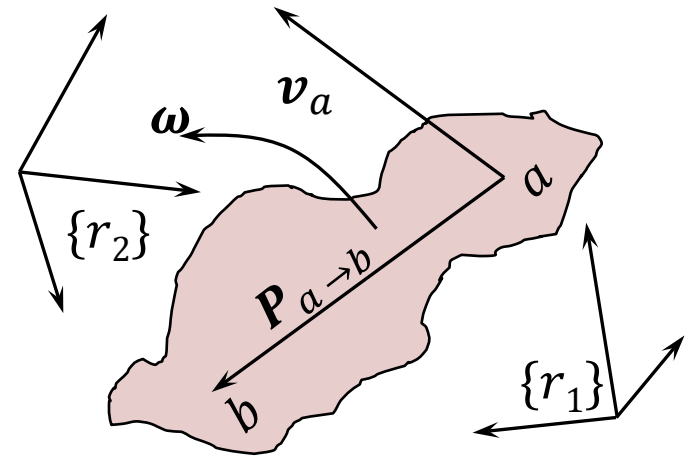
where

$$ {}^{0}\mathbf{J}_{w} = \begin{bmatrix} A_{3\times3} & 0_{3\times3} \\ B_{3\times3} & C_{3\times3} \end{bmatrix} $$

The only drawback is that the joint directions require finding the homogeneous transforms ${}^{0}_{i}\boldsymbol{T}$ which is still tedious. As an alternative, let us consider a Jacobian based on a point at the wrist out oriented as one the intermediate frames $\{3 \text{ or } 4\}$.

Velocity Transformation ${}^{r_2}_{r_1}\boldsymbol{T}\boldsymbol{v}_{b\to a}$

Assume two points "a" and "b" on the same rigid body, let the velocity of point a be known with respect to a reference frame $\{r_1\}$. What is the velocity of point b with respect to frame $\{r_2\}$?



Known: ${}^{r_1}\boldsymbol{V}_a = \begin{bmatrix} {}^{r_1}\boldsymbol{v}_a \\ {}^{r_1}\boldsymbol{\omega} \end{bmatrix}$, find ${}^{r_2}\boldsymbol{V}_b = \begin{bmatrix} {}^{r_2}\boldsymbol{v}_b \\ {}^{r_2}\boldsymbol{\omega} \end{bmatrix}$

$$
\begin{aligned}
{}^{r_2}\boldsymbol{v}_b &= {}^{r_2}_{r_1}\boldsymbol{R}\left({}^{r_1}\boldsymbol{v}_a + {}^{r_1}\boldsymbol{\omega} \times {}^{r_1}\boldsymbol{P}_{a\to b}\right) \\
&= {}^{r_2}_{r_1}\boldsymbol{R}\,{}^{r_1}\boldsymbol{v}_a + {}^{r_2}_{r_1}\boldsymbol{R}\,{}^{r_1}\boldsymbol{P}_{b\to a} \times {}^{r_2}_{r_1}\boldsymbol{R}\,{}^{r_1}\boldsymbol{\omega} \\
&= {}^{r_2}_{r_1}\boldsymbol{R}\,{}^{r_1}\boldsymbol{v}_a + {}^{r_2}\widetilde{\boldsymbol{P}}_{b\to a}\,{}^{r_2}_{r_1}\boldsymbol{R}\,{}^{r_1}\boldsymbol{\omega}
\end{aligned}
$$

$$
{}^{r_2}\boldsymbol{\omega} = {}^{r_2}_{r_1}\boldsymbol{R}\,{}^{r_1}\boldsymbol{\omega}
$$

where the skew matrix ${}^{r_2}\widetilde{\boldsymbol{P}}_{b\to a}$ is: ${}^{r_2}\widetilde{\boldsymbol{P}}_{b\to a} = \begin{bmatrix} 0 & -P_{b\to a_z} & P_{b\to a_y} \\ P_{b\to a_z} & 0 & -P_{b\to a_x} \\ -P_{b\to a_y} & P_{b\to a_x} & 0 \end{bmatrix}$

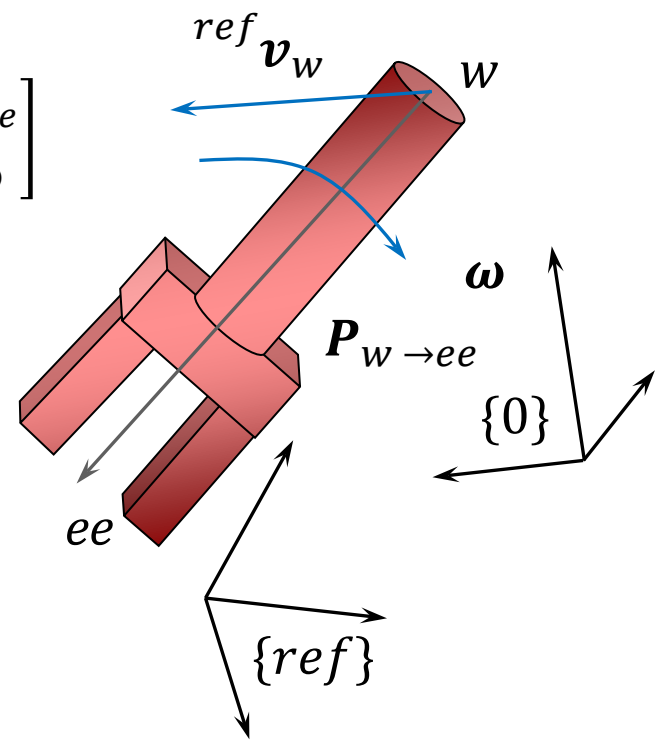
$$
\begin{bmatrix} {}^{r_2}\boldsymbol{v}_b \\ {}^{r_2}\boldsymbol{\omega} \end{bmatrix} = \underbrace{\begin{bmatrix} {}^{r_2}_{r_1}\boldsymbol{R} & {}^{r_2}\widetilde{\boldsymbol{P}}_{b\to a}\,{}^{r_2}_{r_1}\boldsymbol{R} \\ \boldsymbol{0}_{3\times 3} & {}^{r_2}_{r_1}\boldsymbol{R} \end{bmatrix}}_{{}^{r_2}_{r_1}\boldsymbol{T}\boldsymbol{v}_{b\to a}} \begin{bmatrix} {}^{r_1}\boldsymbol{v}_a \\ {}^{r_1}\boldsymbol{\omega} \end{bmatrix}
$$

Velocity Transformation Matrix

5.7 Jacobians – Remarks – Velocity Transformation ${}^{0}_{ref}\boldsymbol{T}\boldsymbol{v}_{ee\to w}$

Let ${}^{ref}\mathbf{J}_w$ be known (next slides) for a given set of joint rates $\dot{\boldsymbol{q}}$. The velocity of the wrist, ${}^{ref}\boldsymbol{V}_w$, can be found. The general velocity transformation can be implemented to the end-effector to find ${}^{0}\boldsymbol{V}_{ee}$.

Known: ${}^{ref}\boldsymbol{V}_w = \begin{bmatrix} {}^{ref}\boldsymbol{v}_w \\ {}^{ref}\boldsymbol{\omega} \end{bmatrix} = {}^{ref}\mathbf{J}_w\dot{\boldsymbol{q}}$, find ${}^{0}\boldsymbol{V}_{ee} = \begin{bmatrix} {}^{0}\boldsymbol{v}_{ee} \\ {}^{0}\boldsymbol{\omega} \end{bmatrix}$



$$
\begin{aligned}
{}^{0}\boldsymbol{v}_{ee} &= {}^{0}_{ref}\boldsymbol{R}\left({}^{ref}\boldsymbol{v}_w + {}^{ref}\boldsymbol{\omega} \times {}^{ref}\boldsymbol{P}_{w\to ee}\right) \\
&= {}^{0}_{ref}\boldsymbol{R}\,{}^{ref}\boldsymbol{v}_w + {}^{0}_{ref}\boldsymbol{R}\,{}^{ref}\boldsymbol{P}_{ee\to w} \times {}^{0}_{ref}\boldsymbol{R}\,{}^{ref}\boldsymbol{\omega} \\
&= {}^{0}_{ref}\boldsymbol{R}\,{}^{ref}\boldsymbol{v}_w + {}^{0}\widetilde{\boldsymbol{P}}_{ee\to w}\,{}^{0}_{ref}\boldsymbol{R}\,{}^{ref}\boldsymbol{\omega} \\
{}^{0}\boldsymbol{\omega} &= {}^{0}_{ref}\boldsymbol{R}\,{}^{ref}\boldsymbol{\omega}
\end{aligned}
$$

where the skew matrix $\widetilde{\boldsymbol{P}}_{ee\to w}$ is

$$
\widetilde{\boldsymbol{P}}_{ee\to w} = \begin{bmatrix} 0 & -P_{ee\to w_z} & P_{ee\to w_y} \\ P_{ee\to w_z} & 0 & -P_{ee\to w_x} \\ -P_{ee\to w_y} & P_{ee\to w_x} & 0 \end{bmatrix} \text{ and } \begin{bmatrix} {}^{0}\boldsymbol{v}_{ee} \\ {}^{0}\boldsymbol{\omega} \end{bmatrix} = \underbrace{\begin{bmatrix} {}^{0}_{ref}\boldsymbol{R} & {}^{0}\widetilde{\boldsymbol{P}}_{ee\to w}\,{}^{0}_{ref}\boldsymbol{R} \\ \mathbf{0}_{3\times 3} & {}^{0}_{ref}\boldsymbol{R} \end{bmatrix}}_{{}^{0}_{ref}\boldsymbol{T}\boldsymbol{v}_{ee\to w}} \begin{bmatrix} {}^{ref}\boldsymbol{v}_w \\ {}^{ref}\boldsymbol{\omega} \end{bmatrix}
$$

Velocity Transformation Matrix

5.7 Jacobians – Remarks – Jacobian $({}^{ref}\mathbf{J}_w)$

In order to determine the Jacobian based on a point on the end-effector coincident with the wrist and oriented with respect to any reference frame, we must use the cross-product method;

$$
{}^{ref}\mathbf{J}_w = \begin{array}{c} \\ \\ \end{array}
\begin{array}{c}
\begin{bmatrix} \cdots & \underbrace{\begin{matrix} {}^{ref}\hat{Z}_i \times {}^{ref}\boldsymbol{P}_{i\to w} \\ {}^{ref}\hat{Z}_i \end{matrix}}_{\text{Revolute Joint } i} & \cdots & \underbrace{\begin{matrix} {}^{ref}\hat{Z}_j \\ 0_{3\times 1} \end{matrix}}_{\text{Prismatic Joint } j} & \cdots \end{bmatrix}
\end{array}
\begin{array}{l} \text{Linear Velocity} \\ \\ \text{Angular Velocity} \end{array}
$$

- The joint direction ${}^{ref}\hat{Z}_i$ can be obtained by extracting the third column to the homogeneous transforms ${}^{ref}_{i}\boldsymbol{T}$.
- The position vector ${}^{i}\boldsymbol{P}_{i\to w}$ is extracted from the last column of ${}^{i}_{w}\boldsymbol{T}$, then ${}^{ref}\boldsymbol{P}_{i\to w}$ $= {}^{ref}_{i}\boldsymbol{R}\ {}^{i}\boldsymbol{P}_{i\to w}$

Inverse Velocity Problem

In the previous sections, our goal was to find the velocity of the end-effector ${}^{0}\boldsymbol{V}_{ee}$ as a function of the joint rates $\dot{\boldsymbol{q}}$. The inverse velocity problem is the opposite, ${}^{0}\boldsymbol{V}_{ee}$ is known and $\dot{\boldsymbol{q}}$ unknown.

First, we invert the velocity transformation matrix or find the appropriate one using the general method shown before:

$$
\begin{aligned}
{}^{ref}\boldsymbol{V}_{w} &= \left({}^{0}_{ref}\boldsymbol{T}\boldsymbol{v}_{ee\,\to w}\right)^{-1}\,{}^{0}\boldsymbol{V}_{ee} \\
&= {}^{ref}_{0}\boldsymbol{T}\boldsymbol{v}_{w\,\to ee}\,{}^{0}\boldsymbol{V}_{ee}
\end{aligned}
$$

Then, $\dot{\boldsymbol{q}}$ is found by inverting the Jacobian matrix

$$
\dot{\boldsymbol{q}} = \left({}^{ref}\mathbf{J}_{w}\right)^{-1}\,{}^{ref}\boldsymbol{V}_{w}
$$

where

$$
\left({}^{ref}\mathbf{J}_{w}\right)^{-1} = \begin{bmatrix} A^{-1} & 0_{3\times3} \\ -C^{-1}BA^{-1} & C^{-1} \end{bmatrix} \quad \text{with} \quad {}^{ref}\mathbf{J}_{w} = \begin{bmatrix} A_{3\times3} & 0_{3\times3} \\ B_{3\times3} & C_{3\times3} \end{bmatrix}
$$

5.8 Singularities

Given that we have a linear transformation relating joint velocity to Cartesian velocity, a reasonable question to ask is: Is this matrix invertible? That is, is it nonsingular? If the matrix is nonsingular, then we can invert it to calculate joint rates from given Cartesian velocities:

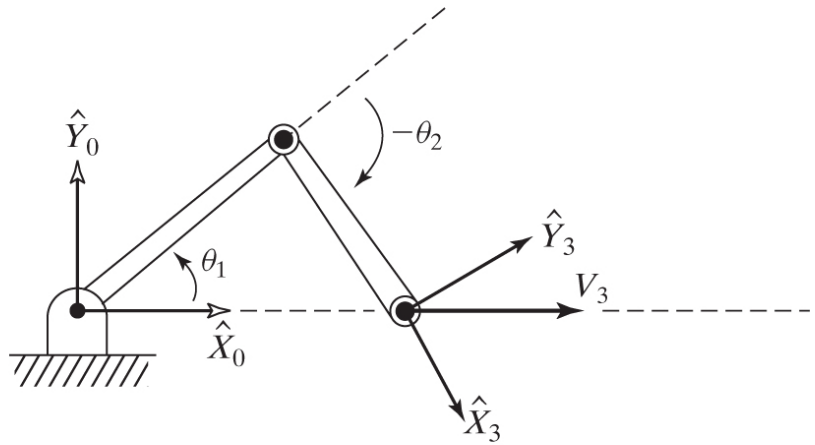
$$\dot{\boldsymbol{q}} = \boldsymbol{J}^{-1}\boldsymbol{v}$$

Singularities occur when the Jacobian becomes singular (rank deficient). Physically, the manipulator loses one or more DoF, i.e. it cannot move in some directions in the end-effector space no matter what joint rates are selected.

Most manipulators have configurations where the Jacobian becomes singular. Such locations are called singularities of the mechanism or singularities for short. In singular points, the danger in applying the inverse Jacobian in a robot control system is that the inverse Jacobian blows up! This results in joint rates approaching infinity as the singularity is approached.

5.8 Singularities

In the previous example, Consider the end-effector is moving along the X axis at 1.0 m/s.



We can find the joint velocities along this trajectory using the inverse Jacobian as:

$$ {}^{0}J_3^{-1} = \frac{1}{L_1 L_2 s_1}\begin{bmatrix} L_2 c_{12} & L_2 s_{12} \\ -L_1 c_1 - L_2 c_{12} & -L_1 s_1 - L_2 s_{12} \end{bmatrix} $$

Then, using the inverse Jacobian equation for a velocity of 1 m/s in the X direction, we can calculate joint rates as a function of manipulator configuration

$$ \dot{\theta}_1 = \frac{c_{12}}{L_1 s_1}, \dot{\theta}_2 = -\frac{c_1}{L_2 s_2} - \frac{c_{12}}{L_1 s_2} $$

as the arm stretches out toward $\theta_2 = 0$, both joint rates go to infinity.

5.8 Singularities

Regardless of the Jacobian ${}^{0}\mathbf{J}_{e.e}$, ${}^{0}\mathbf{J}_{w}$, or ${}^{ref}\mathbf{J}_{w}$, the conditions that make the manipulator be in a singular configuration are the same. Thus, the most convenient Jacobian is ${}^{ref}\mathbf{J}_{w}$. The conditions are found by setting the determinant of the Jacobian to zero, i.e.,

$$\left|{}^{ref}\mathbf{J}_{w}\right| = |A_{3\times3}||C_{3\times3}| = 0 \qquad \text{where} \qquad {}^{ref}\mathbf{J}_{w} = \begin{bmatrix} A_{3\times3} & 0_{3\times3} \\ B_{3\times3} & C_{3\times3} \end{bmatrix}$$

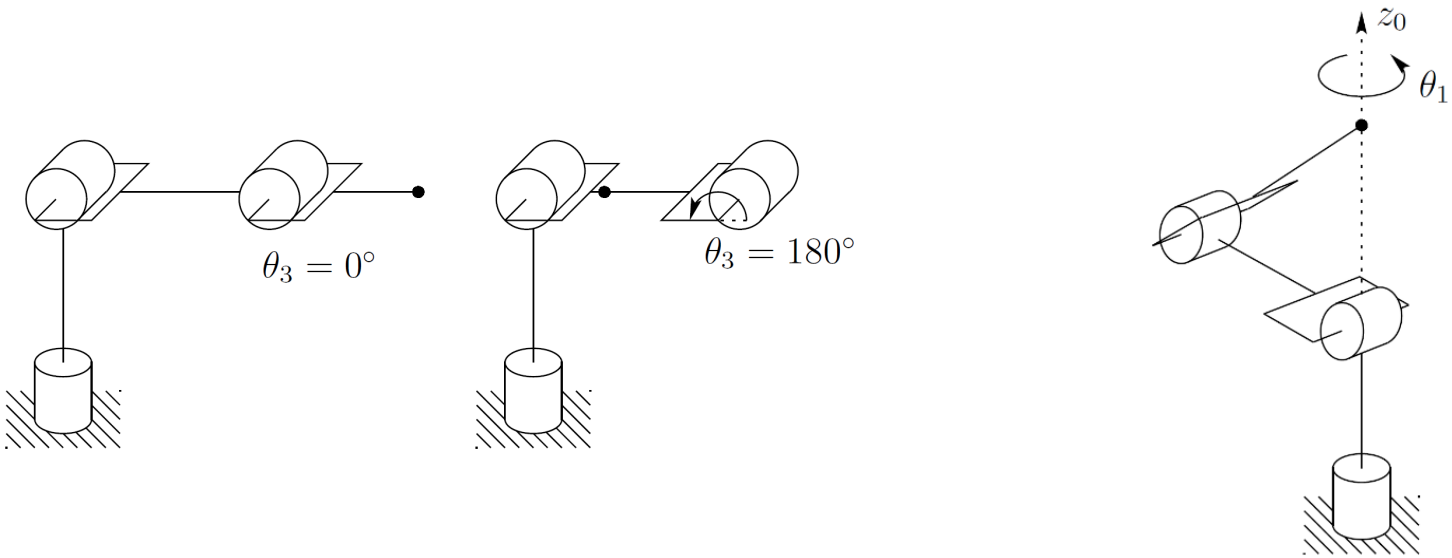
Notice that the determinant of ${}^{ref}\mathbf{J}_{w}$ involves two matrices, $A_{3\times3}$ which is associated to the contribution of the main arm to the linear velocity and $C_{3\times3}$ which is associated to the contribution of the spherical wrist to the angular velocity.

Main Arm Singularities

This singularity occurs when $|A_{3\times3}| = 0$. There are two types:

Boundary singularity: It occurs when the end-effector is on the surface of the workspace (fully stretched or folded back)

Interior Singularities occur within the workspace and depends on the architecture (e.g., two or more axes lined up on a straight line or four revolute joint axes are parallel or intersect at a common point).

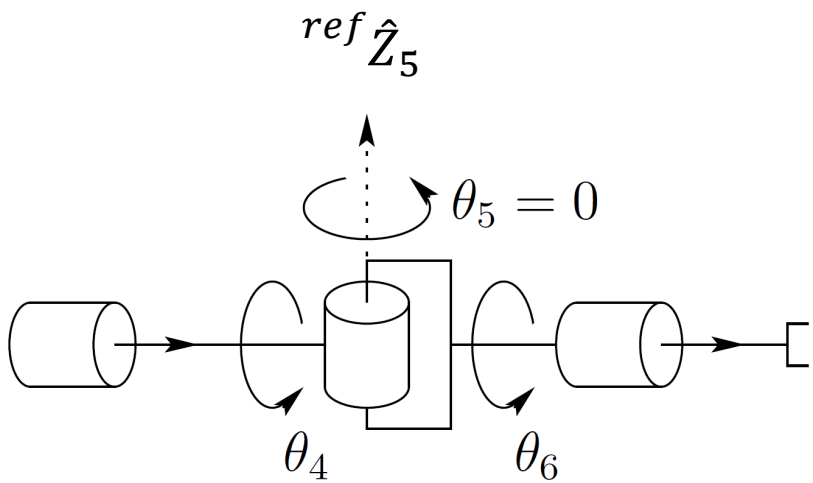


(Boundary and Interior Singularities form Robot Dynamics and Control, 2004)

Wrist Singularities

This singularity occurs when $|C_{3\times3}| = 0$

Since $C_{3\times3} = \left[{}^{ref}\hat{Z}_4 \quad {}^{ref}\hat{Z}_5 \quad {}^{ref}\hat{Z}_6\right]$, a spherical wrist is in a singular configuration whenever the vectors ${}^{ref}\hat{Z}_4$, ${}^{ref}\hat{Z}_5$ and ${}^{ref}\hat{Z}_6$ are linearly dependent

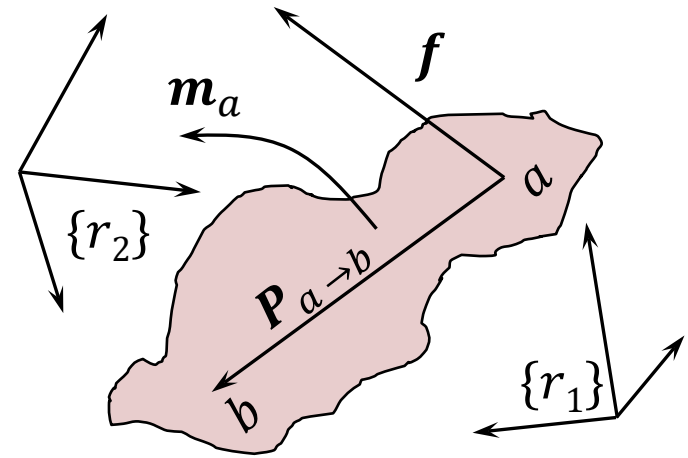


(Wrist Singularities form Robot Dynamics and Control, 2004)

5.9 Static Forces in Manipulators – ${}^{r_2}_{r_1}\boldsymbol{T}_{F_{b\to a}}$

Assume the force and moment about point a be known with respect to a reference frame $\{r_1\}$. What is the force and moment about point b with respect to frame $\{r_2\}$?

Known: ${}^{r_1}\boldsymbol{F}_a = \begin{bmatrix} {}^{r_1}\boldsymbol{f} \\ {}^{r_1}\boldsymbol{m}_a \end{bmatrix}$, find ${}^{r_2}\boldsymbol{F}_b = \begin{bmatrix} {}^{r_2}\boldsymbol{f} \\ {}^{r_2}\boldsymbol{m}_b \end{bmatrix}$



$$
\begin{aligned}
{}^{r_2}\boldsymbol{f} &= {}^{r_2}_{r_1}\mathbf{R}\,{}^{r_1}\boldsymbol{f} \\
{}^{r_2}\boldsymbol{m}_b &= {}^{r_2}_{r_1}\boldsymbol{R}\left({}^{r_1}\boldsymbol{m}_a + {}^{r_1}\boldsymbol{f} \times {}^{r_1}\boldsymbol{P}_{a\to b}\right) \\
&= {}^{r_2}_{r_1}\boldsymbol{R}\,{}^{r_1}\boldsymbol{m}_a + {}^{r_2}_{r_1}\boldsymbol{R}\,{}^{r_1}\boldsymbol{P}_{b\to a} \times {}^{r_2}_{r_1}\boldsymbol{R}\,{}^{r_1}\boldsymbol{f} \\
&= {}^{r_2}_{r_1}\boldsymbol{R}\,{}^{r_1}\boldsymbol{m}_a + {}^{r_2}\widetilde{\boldsymbol{P}}_{b\to a}\,{}^{r_2}_{r_1}\boldsymbol{R}\,{}^{r_1}\boldsymbol{f}
\end{aligned}
$$

where the skew matrix ${}^{r_2}\widetilde{\boldsymbol{P}}_{b\to a}$ is: ${}^{r_2}\widetilde{\mathbf{P}}_{b\to a} = \begin{bmatrix} 0 & -P_{b\to a_z} & P_{b\to a_y} \\ P_{b\to a_z} & 0 & -P_{b\to a_x} \\ -P_{b\to a_y} & P_{b\to a_x} & 0 \end{bmatrix}$

$$
\begin{bmatrix} {}^{r_2}\boldsymbol{f} \\ {}^{r_2}\boldsymbol{m}_b \end{bmatrix} = \underbrace{\begin{bmatrix} {}^{r_2}_{r_1}\boldsymbol{R} & \mathbf{0}_{3\times 3} \\ {}^{r_2}\widetilde{\boldsymbol{P}}_{b\to a}\,{}^{r_2}_{r_1}\boldsymbol{R} & {}^{r_2}_{r_1}\boldsymbol{R} \end{bmatrix}}_{{}^{r_2}_{r_1}\boldsymbol{T}_{F_{b\to a}}} \begin{bmatrix} {}^{r_1}\boldsymbol{f} \\ {}^{r_1}\boldsymbol{m}_a \end{bmatrix}
$$

Force Transformation Matrix

Inverse Static Force Problem

The Inverse Static Force is solved from the equation of conservation of power.

$$\text{Power}_{IN} = \text{Power}_{OUT}$$

$$\boldsymbol{\tau}^{\top}\dot{\boldsymbol{q}} = {}^{0}\boldsymbol{F}_{ee}{}^{\top}\ {}^{0}\boldsymbol{V}_{ee}$$

Since ${}^{0}\boldsymbol{V}_{ee} = {}^{0}\mathbf{J}_{ee}\,\dot{\boldsymbol{q}}$

$$\boldsymbol{\tau}^{\top}\dot{\boldsymbol{q}} = {}^{0}\boldsymbol{F}_{ee}{}^{\top}\ {}^{0}\mathbf{J}_{ee}\,\dot{\boldsymbol{q}}$$

This is true for all $\dot{\boldsymbol{q}}$,

$$\boldsymbol{\tau}^{\top} = {}^{0}\boldsymbol{F}_{ee}{}^{\top}\ {}^{0}\mathbf{J}_{ee}$$

Transposing,

$$\boldsymbol{\tau} = {}^{0}\mathbf{J}_{ee}{}^{\top}\ {}^{0}\boldsymbol{F}_{ee} = {}^{0}\mathbf{J}_{w}{}^{\top}\ {}^{0}\boldsymbol{F}_{w} = {}^{ref}\mathbf{J}_{w}{}^{\top}\ {}^{ref}\boldsymbol{F}_{w}$$

Let The Learning Continue